

8 (a) (i) $a_x = [0.1 + 0.85(1.0 - 0.2)].g = 7.65 = 7.7 \text{ m s}^{-2}$

(ii) Using $v^2 = u^2 + 2as$,

$$0 = \left(\frac{50000}{3600}\right)^2 + 2(-7.65)s$$

$$s = 12.61 = 12.6 \text{ m}$$

(ii) The minimum deceleration will decrease as wet surfaces reduce the frictional force acting on the tyre.

Hence, the braking distance will increase.

- (iv) • Driver and passengers may experience discomfort in travelling in the vehicle.
• Tailgating vehicles may not brake in time and may result in collision.
• Tyre will wear out faster.

(b) (i)

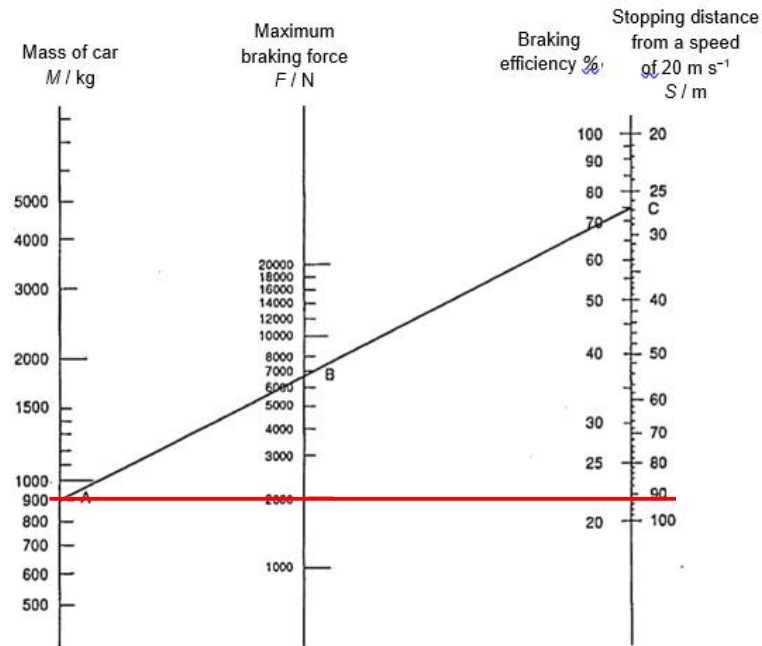


Fig. 8.1

(ii) 22 % , [or 22.5%]

(iii) $22 = (\text{deceleration}/9.81) \times 100$

$$\text{deceleration} = 2.16 \text{ m s}^{-2}$$

or

$$F = ma$$

$$2000 = 900 a$$

$$a = 2.22 \text{ m s}^{-2}$$

(iv) Using $F = ma$,

$$6700 + 2000 = (900)a$$

$$a = 9.67 \text{ m s}^{-2} = 9.7 \text{ m s}^{-2}$$

(v) Not advisable.

- The driver needs to use both his hands to control the steering wheel when braking at high speed. Hence, he should not use his hand to activate the handbrake.
- The car may skid as the car is not designed for both foot brake and hand brake to be used at the same time.

(c) (i) 38.0 m [allow 37.0 m to 39.0 m]

(ii) Point plotted correctly with best fit line drawn

(iii) Explain that work done is product of force and displacement coordinates in the graph. The coordinate is (9000, 29)

$$\text{Work done} = 9000 \times 29 = 2.61 \times 10^5 \text{ J}$$

[to present answer in standard form]

Assumption: the braking force is constant throughout the deceleration.

(d) (i) Using Fig. 8.1, the braking efficiency is 87.5% for braking force of 12000 N. [allow 85% to 90%]

$$\text{Hence, the deceleration of car} = 0.875 \times 9.81 = 8.58 \text{ m s}^{-2}$$

$$\begin{aligned}\text{Force} &= ma \\ &= (70)(8.58) \\ &= 600 \text{ N}\end{aligned}$$

(ii) $v = u + at$,
 $0 = 20 + (-8.58)t$
 $t = 2.3 \text{ s}$

The reaction time for the passengers could be longer than 2.3 s as he was sleeping and would not be able to react quickly when the car was braking.

OR

The force to restrain the passenger could be too large at 600N.